

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q17-0828

Lcec No. 5 138 kV  
April 16, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0828. The Interconnection Customer proposes a 103.1 MW battery energy storage facility interconnecting at the Lcec No. 5 138 kV substation in the West Virginia area. The request was received on July 18, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 275 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 50 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$90,869,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                              | Contingency                                             | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------------------------|---------------------------------------------------------|-------------|--------------|----------|
| Low Gap – Canovanas Pueblo 2401 138 kV                          | Artesia Country Club – Air Force Base 345 kV            | 159.0       | 313          | 5.3      |
| Cheatham – Canovanas Pueblo 2401 138 kV L                       | Air Force Base – Artesia Country Club 345 kV            | 157.9       | 290          | 5.3      |
| Arvin Substation – Shuffleton 138 kV                            | Kearneyville – Clark County 345 kV                      | 141.0       | 223          | 13.7     |
| Arvin – Garden Corners 138 kV L                                 | Douglas City – Southwest Research Institute 138 kV line | 131.2       | 199          | 12.0     |
| Arvin – Southwest Research Institute 138 kV                     | Garden Corners – Douglas City 138 kV                    | 131.2       | 226          | 13.3     |
| Jack McDonough – Low Gap 138 kV                                 | Artesia Country Club – Clark County 345 kV              | 128.7       | 592          | 7.9      |
| Firstenergy R Paul Smith Power Sta – Dam No. 4 Hydro Sta 138 kV | Dam No. 5 Hydro Sta – Lascassas 138 kV                  | 128.2       | 313          | 21.1     |
| Nebraska City 1 – Mud Lake 138 kV                               | Prospect Cenral – Marvin Griffin Rd 138 kV Line         | 127.7       | 237          | 7.0      |
| Air Force Base Substation 345-138 kV Auto                       | Tiger ay – Air Force Base 345 kV L                      | 127.7       | 541          | 7.0      |
| Shepherdstown – Lcec No. 5 138 kV                               | American – Urban 345 kV                                 | 126.1       | 426          | 42.2     |
| Peachland – Sun City 138 kV                                     | Kearneyville – Clark County 345 kV                      | 126.0       | 355          | 17.2     |
| Douglas City – Southwest Reserch Institute 138 kV Line          | Douglas City Substation – Garden Corners 138 kV         | 125.5       | 261          | 13.3     |

| Monitored Facility                               | Contingency                                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------|---------------------------------------------------------------|-------------|--------------|----------|
| Islamorada – Prospect Central Substation 138 kV  | American – Urban 345 kV                                       | 125.0       | 294          | 5.9      |
| Gene – Grimmway-Malaga 138 kV                    | Killingsworth – Gallegos 138 kV                               | 124.1       | 314          | 16.5     |
| Garden Corners – Douglas City 138 kV             | Douglas City – Southwest Research Institute 138 kV            | 123.5       | 241          | 12.0     |
| Gene – Tanner 138 kV                             | Gallegos – Killingsworth Substation 138 kV                    | 123.2       | 327          | 16.5     |
| Clawson – Reid 138 kV L                          | W Waynesboro Substation – Pineville 345 kV                    | 123.0       | 476          | 6.8      |
| Palogen – Highway 138 kV line                    | W Waynesboro – Pineville 345 kV                               | 122.7       | 511          | 6.8      |
| Carrel – Buffalo Tungsten 138 kV                 | Chambersburg – W Waynesboro 345 kV                            | 122.4       | 485          | 10.0     |
| Buffalo Tungsten – Chevmain 138 kV line          | W Waynesboro – Chambersburg 345 kV                            | 122.4       | 479          | 10.0     |
| Lcec No. 5 – Tanner 138 kV                       | Killingsworth – Gallegos 138 kV                               | 122.3       | 336          | 16.5     |
| Baraboo – Nebraska City 1 138 kV L               | Prospect Central – Marvin Griffin Road 138 kV                 | 122.2       | 248          | 7.0      |
| Islamorada – St. John 138 kV                     | Urban – American 345 kV L                                     | 121.4       | 299          | 5.9      |
| St. John – Mud Lake 138 kV                       | Prospect Central Substation – Marvin Griffin Road 138 kV Line | 119.9       | 242          | 7.0      |
| Nipetown – Dam No. 5 Hydro Sta 138 kV L          | American – Urban 345 kV                                       | 119.6       | 314          | 9.6      |
| Douglas City – Millville 138 kV                  | Clark County – Kearneyville 345 kV                            | 119.4       | 461          | 16.4     |
| N Mesa – Columbus City 138 kV                    | International Paper Valliant Ok – Kearneyville 345 kV         | 119.2       | 372          | 24.3     |
| Wst Waynesboro – Highway 138 kV                  | Pineville – West Waynesboro 345 kV                            | 119.1       | 541          | 6.8      |
| Sleepy Hollow – Millville Substation 138 kV Line | Clark county – Kearneyville 345 kV                            | 119.0       | 474          | 16.4     |
| Claytonia – Shuffleton 138 kV                    | Clark County – Kearneyville 345 kV                            | 118.9       | 241          | 13.7     |
| Sleepy Hollow – Grimmway-Malaga 138 kV           | Clark County – Kearneyville Substation 345 kV                 | 118.6       | 487          | 16.4     |
| Columbus City 138/1 kV Transformer               | International Paper Valliant Ok – Kearneyville 345 kV         | 118.6       | 495          | 23.2     |
| Prospect Central – Marvin Griffin Rd 138 kV      | American – Urban Substation 345 kV Line                       | 117.3       | 314          | 5.9      |
| Reid – Palogen 138 kV L                          | W Waynesboro Substation – Pineville 345 kV L                  | 117.2       | 514          | 6.8      |
| St. John – Rio Dell 138 kV line                  | American – Urban 345 kV Line                                  | 116.4       | 513          | 10.8     |
| Antetam – Clawson Substation 138 kV              | W Waynesboro – Pineville 345 kV                               | 115.7       | 475          | 6.8      |
| University City – Lamont 138 kV                  | Clark County – Kearneyville 345 kV                            | 114.1       | 342          | 17.2     |
| Palmyra – Grimmway-Malaga 138 kV                 | Tanner Substation – LcecNo. 5 138 kV                          | 113.4       | 313          | 7.9      |
| Glenbrook – Palmyra 138 kV Line                  | Tanner – Lcec No. 5 138 kV                                    | 113.1       | 325          | 7.9      |
| Dkalb – Robison Park 230 kV                      | Jones Chapel – Dekalb 500 kV                                  | 112.9       | 1,515        | 6.5      |
| Atietam – Pineville Substation 138 kV            | W Waynesboro – Pineville 345 kV                               | 112.8       | 456          | 6.8      |

| Monitored Facility                                           | Contingency                                 | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------------|---------------------------------------------|-------------|--------------|----------|
| Redman Substation – Glenbrook 138 kV line                    | Tanner – Lcec No. 5 138 kV                  | 112.8       | 336          | 7.9      |
| Redman – Gallegos 138 kV L                                   | Tanner – Lcec No. 5 138 kV                  | 112.6       | 344          | 7.9      |
| Lascassas – Dm No. 5 Hydro Sta 138 kV                        | American – Urban 345 kV                     | 112.5       | 347          | 9.6      |
| Gallegos – Killingsworth 138 kV                              | Tanner – Lcec No. 5 Substation 138 kV L     | 112.4       | 354          | 7.9      |
| Dam No. 4 Hydro Sta – Schafrath Swith Substation 138 kV line | American – Urban 345 kV                     | 111.3       | 317          | 9.6      |
| Chevmain – Cressey 138 kV line                               | W Waynesboro – Chambersburg 345 kV          | 111.1       | 521          | 10.0     |
| Shepherdstown Substation – Gravel Ridge 138 kV L             | American Substation – Urban 345 kV line     | 110.9       | 605          | 26.8     |
| Dam No. 4 Hydro Station – Gravel Ridge 138 kV L              | American – Urban 345 kV line                | 110.8       | 612          | 26.8     |
| Cressy – Firstenergy R Paul Smith Power Station 138 kV L     | Dam No. 5 Hydro Sta – Lascassas 138 kV line | 110.4       | 328          | 21.1     |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Low Gap – Canovanas Pueblo 2401 138 kV                               | Rebuild line with higher-capacity conductor | \$5,296,000   | 14% (\$766,000)       |
| Cheatham – Canovanas Pueblo 2401 138 kV L                            | Rebuild line with higher-capacity conductor | \$2,842,000   | 14% (\$411,000)       |
| Arvin – Shuffleton 138 kV                                            | Rebuild line with higher-capacity conductor | \$4,289,000   | 38% (\$1,646,000)     |
| Arvin – Garden Corners 138 kV L                                      | Rebuild line with higher-capacity conductor | \$13,882,000  | 37% (\$5,198,000)     |
| Arvin – Southwest Research Institute 138 kV                          | Rebuild line with higher-capacity conductor | \$8,685,000   | 37% (\$3,252,000)     |
| Jack McDonough – Low Gap 138 kV                                      | Rebuild line with higher-capacity conductor | \$1,839,000   | 6% (\$104,000)        |
| Dam No. 4 Hydro Sta – Firstenergy R Paul Smith Power Sta 138 kV line | Rebuild line with higher-capacity conductor | \$3,054,000   | 11% (\$335,000)       |
| Nebraska City 1 – Mud Lake 138 kV                                    | Rebuild line with higher-capacity conductor | \$13,456,000  | 47% (\$6,383,000)     |
| Air Force Base 345/138 kV XFMR                                       | Replace transformer with higher-rated unit  | \$10,185,000  | 21% (\$2,163,000)     |
| Shepherdstown – Lcec No. 5 138 kV                                    | Rebuild line with higher-capacity conductor | \$3,234,000   | 27% (\$868,000)       |
| Peachland – Sun City 138 kV                                          | Rebuild line with higher-capacity conductor | \$5,881,000   | 28% (\$1,618,000)     |

| Facility                                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Douglas City – Southwest Research Institute 138 kV Line | Rebuild line with higher-capacity conductor | \$3,917,000   | 37% (\$1,467,000)     |
| Islamorada – Prospect Central 138 kV                    | Rebuild line with higher-capacity conductor | \$4,417,000   | 100% (\$4,417,000)    |
| Gene – Grimmway-Malaga 138 kV line                      | Rebuild line with higher-capacity conductor | \$10,753,000  | 39% (\$4,167,000)     |
| Garden Corners – Douglas City 138 kV                    | Rebuild line with higher-capacity conductor | \$2,521,000   | 37% (\$944,000)       |
| Tanner – Gene Substation 138 kV                         | Rebuild line with higher-capacity conductor | \$679,000     | 39% (\$263,000)       |
| Clawson – Reid 138 kV L                                 | Rebuild line with higher-capacity conductor | \$5,790,000   | 4% (\$229,000)        |
| Palogen – Highway 138 kV line                           | Rebuild line with higher-capacity conductor | \$5,368,000   | 4% (\$212,000)        |
| Buffalo Tungsten – Carrel 138 kV                        | Rebuild line with higher-capacity conductor | \$1,812,000   | 6% (\$101,000)        |
| Buffalo Tungsten – Chevmain 138 kV                      | Rebuild line with higher-capacity conductor | \$5,395,000   | 6% (\$299,000)        |
| Tanner – Lcec No. 5 138 kV line                         | Rebuild line with higher-capacity conductor | \$7,979,000   | 39% (\$3,092,000)     |
| Baraboo – Nebraska City 1 138 kV L                      | Rebuild line with higher-capacity conductor | \$20,941,000  | 47% (\$9,934,000)     |
| Islamorada – St. John 138 kV                            | Rebuild line with higher-capacity conductor | \$6,414,000   | 100% (\$6,414,000)    |
| St. John – Mud Lake 138 kV L                            | Rebuild line with higher-capacity conductor | \$9,900,000   | 47% (\$4,697,000)     |
| Nipetown – Dam No. 5 Hydro Sta 138 kV                   | Rebuild line with higher-capacity conductor | \$3,633,000   | 28% (\$1,004,000)     |
| Douglas City – Millville 138 kV                         | Rebuild line with higher-capacity conductor | \$12,755,000  | 38% (\$4,896,000)     |
| Columbus City – North Mesa 138 kV L                     | Rebuild line with higher-capacity conductor | \$1,603,000   | 44% (\$705,000)       |
| Wst Waynesboro – Highway 138 kV                         | Rebuild line with higher-capacity conductor | \$1,171,000   | 4% (\$46,000)         |
| Sleepy Hollow – Millville Substation 138 kV Line        | Rebuild line with higher-capacity conductor | \$6,683,000   | 38% (\$2,565,000)     |
| Claytonia – Shuffleton 138 kV                           | Rebuild line with higher-capacity conductor | \$2,507,000   | 38% (\$962,000)       |
| Sleepy Hollow Substation – Grimmway-Malaga 138 kV line  | Rebuild line with higher-capacity conductor | \$2,222,000   | 38% (\$853,000)       |
| Columbus City 138/1 kV Transformer                      | Replace transformer with higher-rated unit  | \$5,129,000   | 18% (\$901,000)       |
| Prospect Central – Marvin Griffin Rd 138 kV             | Rebuild line with higher-capacity conductor | \$5,930,000   | 100% (\$5,930,000)    |
| Reid – Palogen 138 kV                                   | Rebuild line with higher-capacity conductor | \$518,000     | 4% (\$20,000)         |
| St. John – Rio Dell 138 kV line                         | Rebuild line with higher-capacity conductor | \$7,041,000   | 8% (\$588,000)        |

| Facility                                                 | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Antetam – Clawson Substation 138 kV                      | Rebuild line with higher-capacity conductor | \$4,154,000   | 4% (\$164,000)        |
| University City – Lamont 138 kV                          | Rebuild line with higher-capacity conductor | \$14,117,000  | 28% (\$3,883,000)     |
| Palmyra – Grimmway-Malaga 138 kV                         | Rebuild line with higher-capacity conductor | \$2,492,000   | 38% (\$959,000)       |
| Palmyra – Glenbrook 138 kV                               | Rebuild line with higher-capacity conductor | \$1,129,000   | 38% (\$434,000)       |
| Dkalb – Robison Park 230 kV                              | Rebuild line with higher-capacity conductor | \$55,367,000  | 1% (\$494,000)        |
| Atietam – Pineville Substation 138 kV                    | Rebuild line with higher-capacity conductor | \$4,019,000   | 4% (\$159,000)        |
| Glenbrook – Redman 138 kV                                | Rebuild line with higher-capacity conductor | \$2,469,000   | 38% (\$950,000)       |
| Redman – Gallegos 138 kV L                               | Rebuild line with higher-capacity conductor | \$2,657,000   | 38% (\$1,023,000)     |
| Dam No. 5 Hydro Sta – Lascassas 138 kV                   | Rebuild line with higher-capacity conductor | \$1,714,000   | 28% (\$474,000)       |
| Gallegos – Killingsworth 138 kV                          | Rebuild line with higher-capacity conductor | \$4,784,000   | 38% (\$1,841,000)     |
| Schafrath Switch – Dam No. 4 Hydro Sta 138 kV            | Rebuild line with higher-capacity conductor | \$2,682,000   | 28% (\$741,000)       |
| Chevmain – Cressey 138 kV L                              | Rebuild line with higher-capacity conductor | \$8,489,000   | 6% (\$471,000)        |
| Shepherdstown – Gravel Ridge 138 kV                      | Rebuild line with higher-capacity conductor | \$1,749,000   | 12% (\$218,000)       |
| Dam No. 4 Hydro Station – Gravel Ridge 138 kV L          | Rebuild line with higher-capacity conductor | \$6,103,000   | 12% (\$762,000)       |
| Cressy – Firstenergy R Paul Smith Power Station 138 kV L | Rebuild line with higher-capacity conductor | \$8,136,000   | 10% (\$846,000)       |
|                                                          | Total allocated to project                  |               | \$90,869,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.